HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 311: Theory of Computing

Batch: 26th
Section: 2
Date: 08/06/2026

Semester: Spring 2026
Time: 1 Hour 30 Minutes
Total Marks: 30

Mid-Term Examination

Answer any five questions from the following. $5 \times 6 = 30$

			CLOs	Marks
1	(a).	Define automata theory.	CLO1	1
	(b).	Consider set A has 'a' elements and set B has 'b' elements. Explain how many elements are in A×B.	CLO1	2
	(c).	Let A be the set {1,2,3} and B be the set {2,3}. Find the followings:	CLO1	3

i. Is 'A' a subset of 'B'?
ii. Is 'B' a subset of 'A'?
iii. What is A ∪ B? $2^2 = 4$
iv. What is A ∩ B?
v. What is A × B?
vi. What is the power set of B?

			CLOs	Marks
2	(a).	What are domain and range of a set?	CLO1	1
	(b).	Let X be the set {1, 2, 3, 4} and Y be the set {5, 6, 7, 8}. The unary function f: X → Y and the binary function g: X × Y → Y are described in the following tables:	CLO1	5

n	$f(n)$
1	5
2	7
3	5
4	8

g	5	6	7	8
1	6	6	6	6
2	5	6	7	8
3	8	7	6	5
4	7	7	8	5

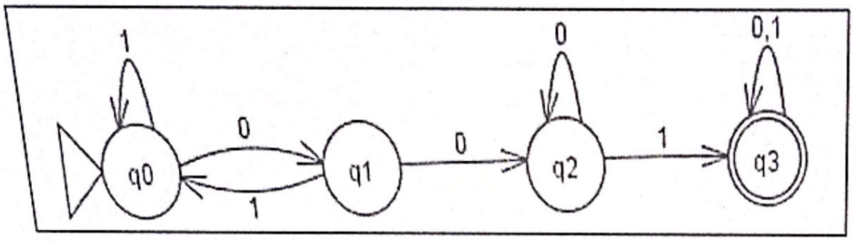
i. What is the value of f(3)?
ii. What are the range and domain of f?
iii. What is the value of g(3, 7)?
iv. What are the range and domain of g?
v. What is the value of g(2, f(4))?

			CLOs	Marks
3	(a).	Briefly explain proof by Induction.	CLO4	2
	(b).	For all n ≥ 1, prove that,	CLO4	4

$$1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

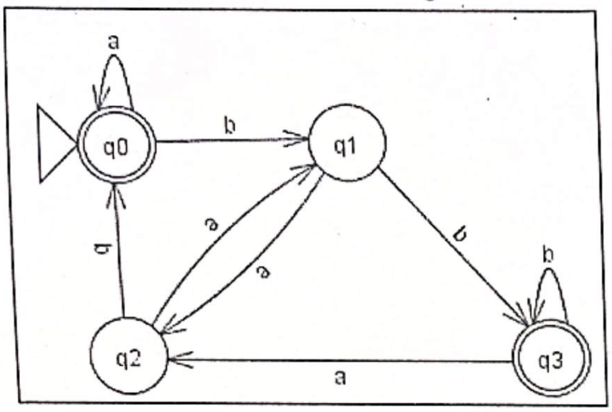
			CLOs	Marks
4	(a).	Briefly explain Deterministic Finite Automata with an example.	CLO3	3

(b). Consider the following DFA design: CLO3 3



Write the regular language for this design with transition table.

5 (a). What is dead state in automata? Consider the given DFA: CLO3 3



i. What is the start state?
ii. What is the set of accept states?
iii. What sequence of states does the machine go through on input 'aabb'?
iv. Does the machine accept the string 'aabb'?
v. Does the machine accept the string ε?

(b). Construct DFA design for the following languages along with its transition table: CLO3 3

i. Construct a DFA over $\sum = \{a, b, c\}$ that accepts only the input string "abca".
ii. Construct a DFA with sigma $\sum = \{0, 1\}$, accepts any string that starts with "1" and ends with "0".

6 (a). Write the formal definition of Non-deterministic Finite Automata (NFA) with an example. CLO2 3

(b). Write the differences between DFA and NFA. CLO2 3

7 (a). Consider the regular language of question 5(b). Now construct the NFA for the given language. CLO3 3

(b). Construct DFA design for the following languages with transition table: CLO3 3

i. {w | w has an odd number of a's and ends with a 'b'}
ii. {w | w ends with 00} with three states

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE311:Theory of Computing

Batch: 26th
Sec: 1
Date: 08/06/2026

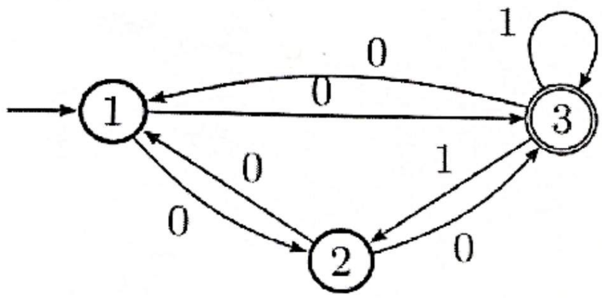
Semester: Spring 2026
Time:1 Hour 30 minutes
Total Marks: 30

Mid-Term Examination

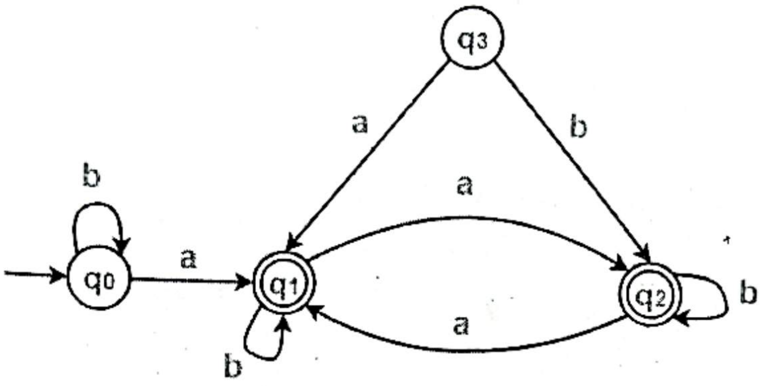
Answer any **five** questions.

Marks: 5×6 = 30

			Marks	CLOS
1.	(a)	Write some applications of automata.	2	1
	(b)	Describe the five tuples of finite automata.	4	2
2.	(a)	Compare between Σ^+ and Σ^*, when Σ={abc}.	2	2
	(b)	Draw an NFA for the string $\mathbf{0^* + (01)^*}$, over the alphabet $\Sigma = \{0,1\}$.	4	3
3.	(a)	With proper diagram, differentiate between NFA and DFA.	3	2
	(b)	Write short notes on the following terms i) Alphabet ii) Power of Alphabet iii) Concatenation of string	3×1=3	2
4.		Design DFA for the following strings: a) Ends with '**abcd**', over the alphabet $\Sigma = \{a,b,c,d\}$. b) (a|b)*caa|b over the alphabet $\Sigma = \{a,b,c\}$.	2×3=6	3
5.		Convert the following NFA to a DFA:	6	3



6.		Design a DFA accepting the set of all strings starts with '**abc**', over the alphabet {a,b,c}. Show that **abcabc** is accepted by the system.	6	3
7.		Using DFA minimization techniques, minimize state(s) of the given DFA.	6	3


q3
a
b
b
a
q0
a
q1
q2
b
a
b

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313 Database Management System

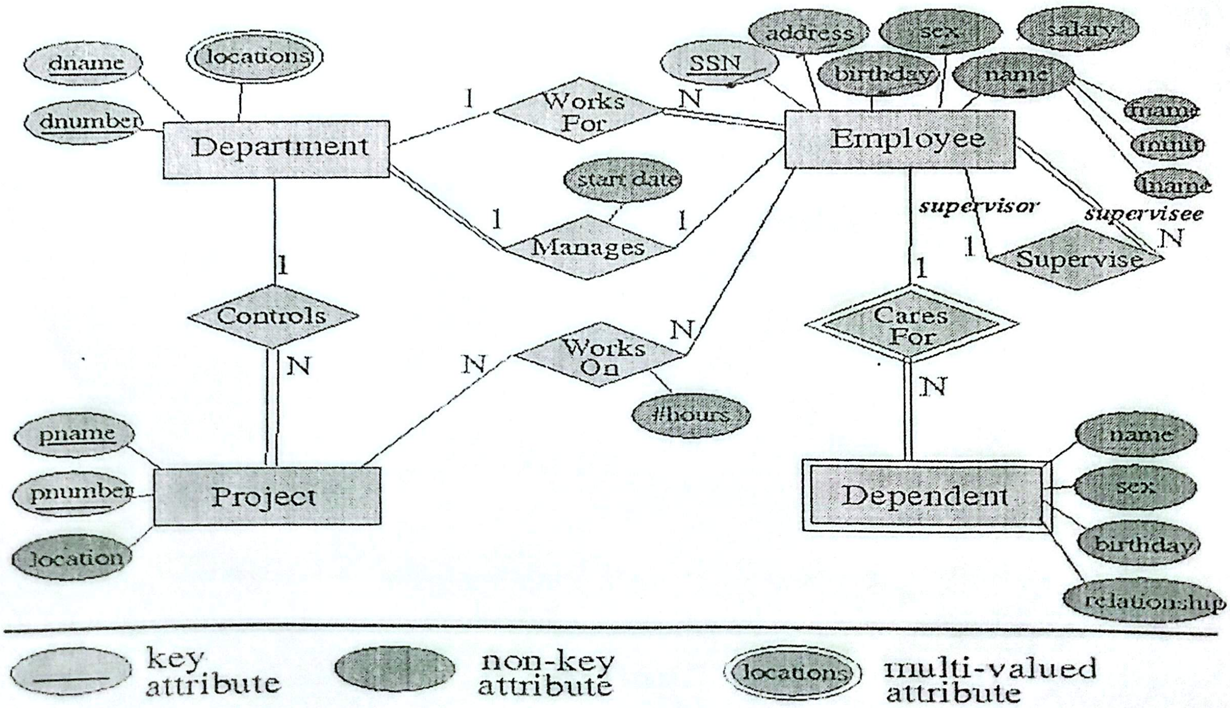
Batch: 26th — Semester: Spring 2026

Date: 10/06/2026 — **Mid Term Examination** — Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions. **5 x 6 = 30**

Q.	Part	Question	CLO	Marks
1.	(a)	Define field. State the components of a database table.	CLO 1	2
	(b)	Explain different types of field in a database.	CLO 1	4
2.	(a)	What is meant by a column key?	CLO 1	1
	(b)	Define different types of keys with examples that are used in a Database Management System.	CLO 1	5
3.	(a)	State the comparison of observation and document analysis of a Database Management System.	CLO 1	3
	(b)	Write the advanced rules that govern data flow diagramming.	CLO 1	3
4.	(a)	Define system analyst.	CLO 1	1
	(b)	Briefly explain prototyping with its advantages and disadvantages.	CLO 1	5
5.	(a)	State the art and guidelines for a successful interview to determine requirements of a DBMS.	CLO 1	2
	(b)	Briefly explain the interview to determine requirements of a DBMS with advantages and disadvantages.	CLO 1	4
6.		Briefly explain the merits and demerits of different types of record based logical model.	CLO 1	6
7.		Consider the following E-R diagram:	CLO 2	6



Write the SQL command to create all the relations. Find the Questionnaires for the given E-R diagram.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 315: Microprocessors & Embedded System

Batch: 26th

Semester: Spring 2026

Date: 13/06/2026 **Mid-Term Examination** Time: 1 Hour 30 minutes

Total Marks: 30

Answer any **five** questions of the following. $5 \times 6 = 30$

1.	(a)	Write difference between Microprocessor and Microcontroller. Explain the memory segmentation of 8086 Microprocessor?	CLO3	3
	(b)	8086 fetches several bytes ahead of execution and stores them in a queue. Explain how the execution unit interacts with this queue, and describe two situations when the prefetch queue is flushed or partially emptied, affecting performance.	CLO3	3
2.	(a)	In 8086, instruction fetching and execution overlap. Explain the benefit of this mechanism with proper figure.	CLO3	3
	(b)	Outline the major differences between Intel Core 2, Intel Core i7, Intel Core i5, Intel Core i3.	CLO3	3
3.	(a)	Draw a simple block diagram of the processor architecture which have highly customized set of instructions and describe four major design features that give it performance advantages over traditional complex-instruction processors.	CLO3	3
	(b)	Compare the older 8-bit architecture with the newer 16-bit architecture in terms of: • Addressing capability. • Instruction execution style. • Pipelining. • Size, I/O and Cost.	CLO3	3
4.	(a)	Explain the following addressing mode with examples: i. Immediate Addressing ii. Implied Addressing iii. Register Addressing	CLO2	3
	(b)	The status indicator in 8086 includes: • *Bits that show arithmetic results* • *Bits that influence program flow* *Categorize these two groups, and explain how different operations set, reset, or use these bits.*	CLO3	3

RISC

5.	(a)	Explain the process of converting Physical address to logical address with diagram.	CLO1	3
	(b)	5(b). Elaborate the following terms with diagram: i. Swapping ii. Fragmentation iii. Partitioning	CLO3	3
6.	(a)	What is memory banking? Explain ODD Bank in details.	CLO3	3
	(b)	6(b). Draw the pin diagram of Memory Banking and Examine the operation for different combination of enable keys (A_0 and $\overline{BHE}$) in ODD and EVEN Bank.	CLO3	3
7.	(a)	Explain Segment Offset Pointers of Microprocessor 8086.	CLO3	3
	(b)	Explain how the microprocessor interprets higher level languages with diagram.	CLO1	3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 317: Computer Networks

Batch: 26^{th}		Semester: Spring 2026
Section: 1 & 2		Time: 1 Hour 30 Minutes
Date: 15/06/2025	**Mid-Term Examination**	Total Marks: 30

Answer any **five** questions from the following. $5 \times 6 = 30$

Q	Part	Question	CLOs	Marks
1	(a).	What is 'node' in computer networks?	CLO1	1
	(b).	Briefly explain the prime attributes of computer networks.	CLO1	2
	(c).	Explain basic types of data used in computer networks.	CLO1	3
2	(a).	Define network topology. Write the names of the basic network topologies.	CLO4	1
	(b).	Briefly explain the basic elements of sender to receiver data transfer procedure.	CLO4	2
	(c).	Define distribution processing. Explain star and mash network topologies with advantages and disadvantages.	CLO4	3
3	(a).	Define protocol in computer networks. Write about the key elements of networking protocols.	CLO4	2
	(b).	Write the names of OSI model layers. Explain the lower layer protocols with functionalities.	CLO5	4
4	(a).	Write the names of IP address classes with priority bits.	CLO2	1
	(b).	Identify following IP address classes: i. 173.150.20.10 ii. 191.120.110.1 iii. 20.20.30.30 iv. 254.255.255.0	CLO2	0.5×2=2
	(c).	What are public and private IP addresses? Write the private IP address ranges for all classes.	CLO2	3
5	(a).	Define default gateway and subnet mask.	CLO2	2
	(b).	Consider an IP address 192.168.0.0/20 containing 4096 IP addresses for a University network configuration. It has **four** computer labs respectively lab1, lab2, lab3 and lab4 that required IP addresses 900, 500, 400, and 60. The network two routers and four switches for respective labs connectivity. Use only one serial cable link line for the connection between two routers. **Draw** the network as described and calculate the variable length subnet mask **(VLSM)** of this	CLO2	4

network.

6	(a).	Briefly explain classless inter-domain routing.	CLO2	2
	(b).	Consider an IP address 192.168.20.0 and subnet mask 255.255.255.248. Find the followings using fixed length subnet mask (FLSM) procedure: i. 1^{st} IP of 3^{rd} block ii. Second last IP of last block iii. 5^{th} valid host of 2^{nd} block iv. Broadcast address of 1^{st} block	CLO2	$1\times4=4$
7	(a).	What is carrier sense multiple access (CSMA)? Explain 1-persistent and Non-persistent CSMA modes.	CLO4	3
	(b).	Write the differences between HUB and Switch.	CLO4	3

HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT317 — Course Title: Engineering Mathematics
Midterm Examination — Semester: Spring 2026
Date: 6/20/2026 — Total Time: 1:30 Hours — Total Marks: 30

Answer any three (03) out of the following questions: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No		Questions	CLOs	Marks
Q1.	(a)	Define Laplace Transform?	CLO1	[1]
	(b)	Let $f(t)$ and $g(t)$ be any function whose Laplace Transform exists. For constant a, b prove: $\mathcal{L}\{a\, f(t) + b\, g(t)\} = a\, \mathcal{L}\{f(t)\} + b\, \mathcal{L}\{g(t)\}$	CLO2	[3]
	(c)	Find the Laplace Transformation of the following functions: (i) $f(t) = e^{at}$ (ii) $f(t) = t$ (iii) $u_a(t) = \begin{cases} 0, & t < a \\ 1, & t > a \end{cases}$	CLO3	[6]
Q2.	(a)	What do you mean by inverse Laplace Transformation?	CLO1	[1]
	(b)	What do you mean by convolution? With the help of convolution find the inverse Laplace Transform of $F(s) = \frac{1}{s(s^2+1)}$	CLO1 CLO3	[4]
	(c)	Solve the following differential equation using Laplace Transformation: $\frac{d^2y}{dt^2} - 2\frac{dy}{dt} - 8y = 0; \quad y(0) = 3,\ y'(0) = 6$	CLO3	[5]
Q3.	(a)	Suppose $f(t)$ is a real valued function whose Laplace Transform is $F(s) = \int_0^\infty e^{-st} f(t)\, dt$. Let the translated function $u_a(t) f(t-a)$ is defined as $u_a(t) f(t-a) = \begin{cases} 0, & 0 < t < a \\ f(t-a), & t > a \end{cases}$ Then prove that, $\mathcal{L}\{u_a(t) f(t-a)\} = e^{-as} F(s)$	CLO2	[4]
	(b)	Solve the following system of differential equations using Laplace Transformation: $\frac{dx}{dt} + y = 3e^{2t}; \quad x(0) = 2$; $\frac{dy}{dt} + x = 0; \quad y(0) = 0$	CLO3	[6]
Q4.	(a)	Define Fourier Series.	CLO1	[1]
	(b)	Answer the followings: (i) What do you mean by orthogonal set of functions? Give an example of an orthogonal set of functions. (ii) What do you mean by orthonormal set of functions?	CLO1	[2]

$-2 \times (-3)$

$\frac{12}{2} = 6B$

$1 \times 3 =$

$6/3 = 2$

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(b) Find the Fourier coefficients of the following series defined on the interval $[-p, p]$. CLO1 [7]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n cos\frac{n\pi x}{p} + b_n sin\frac{n\pi x}{p}\right)$$

Q5. (a) Define half range Fourier sine and cosine series. CLO1 [1]

(b) Find the Fourier cosine series for: CLO3 [5]

$$f(x) = \begin{cases} x^2, & 0 \le x \le 2 \\ 4, & 2 \le x \le 4 \end{cases}$$

(c) Find the Fourier coefficient for: CLO3 [4]

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 \le x < \pi \end{cases}$$